

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed agent intelligence end-to-end — prompts, reasoning, tool calls, and architecture: shipped to production a self-service analytics agent over 67 BigQuery tables with a Context Researcher → SQL Drafter → Validator → Devil's Advocate sub-agent flow, 8 deterministic SQL rules, and human-in-the-loop validation. Approved by Sr. Director and product leadership; currently the only AI skill in active use by business teams — an agent taking real actions in production, not a side project.
- Built and shipped an autonomous Jira agent that takes real actions for the business: 6-gate triage, MCP-driven tool calls / context gathering, agentic RAG (FAISS + BM25) over 11,000 historical SQL queries, and self-healing execution; closed 400+ requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10. Owned it from design to debug to deploy.
- Partnered with Product and business stakeholders to turn business challenges into agentic solutions and drive adoption — consultative work, not just code: built a hybrid orchestrator that lets Product self-serve requests they previously filed as tickets, and evaluate AI outputs with engineering rigor (23-case golden-query suite, regression tests, guardrails).
- Built LLM-powered services in Python on FastAPI + Cloud Run, working across Claude / GPT / Gemini; delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- Built a RAG-to-proposal generator for an enterprise consulting client, deployed on AWS EC2; partnered with the client to scope and ship it end-to-end; cut proposal time ~30%; **Best in Show — UChicago MSADS Capstone.**

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant experiment; controlled A/B tests + prompt engineering persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); partnered with district stakeholders; retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit front-end + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent building: agent intelligence design (prompts, reasoning, tool calls, architecture), sub-agent orchestration, MCP server design & tool integration, agentic workflows, prompt engineering, agent-behavior evaluation & guardrails

LLM products: RAG & vector search (embeddings + FAISS + BM25 + RRF), multi-model routing (Claude / GPT / Gemini), evals & testing (golden sets, regression, human-in-the-loop), context engineering

Client-facing delivery: business→agentic-solution translation, stakeholder partnership & adoption, consulting capstone (client delivery, Best in Show), cross-functional collaboration

Engineering: Python, FastAPI, REST APIs, Cloud Run, GCP / AWS, OOP, CI/CD (pytest), decision logging / observability; JavaScript (working), Streamlit + React Recharts front-ends

Data: BigQuery, dbt, SQL, PySpark, Airflow, Databricks, MySQL

LLMs / dev: Claude (Opus / Sonnet / Haiku), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI; Claude Code, Cursor

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL